

英文学术论文写作指南

第二讲：学术研究之写作思路

于静 副研究员

中国科学院信息工程研究所

2022.07 @ Bilibili



中国科学院 信息工程研究所
INSTITUTE OF INFORMATION ENGINEERING, CAS



中国科学院大学
University of Chinese Academy of Sciences

从科学问题是否探究本质说起

CCF—A

科学问题 聚焦领域前沿，探索本质问题

解决方法 专注核心挑战，具普适性方法

Mucko: Multi-Layer Cross-Modal Knowledge Reasoning for Fact-based Visual Question Answering

Zihao Zhu^{1,2*}, Jing Yu^{1,2*†}, Yujing Wang³, Yajing Sun^{1,2}, Yue Hu^{1,2} and Qi Wu⁴

¹Institute of Information Engineering, Chinese Academy of Sciences, Beijing, China

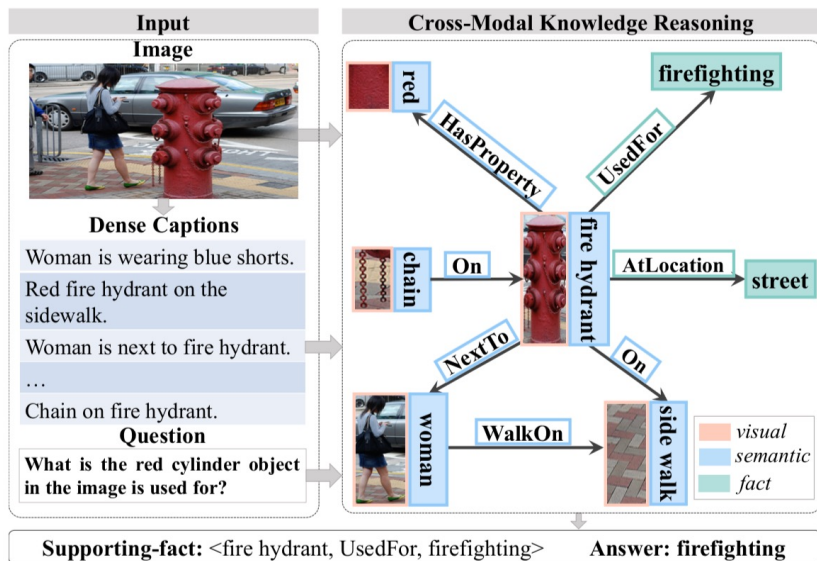
²School of Cyber Security, University of Chinese Academy of Sciences, Beijing, China

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IJCAI 2020

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CCF—C

追逐领域热点，解决表面问题

盲从热门方法，模型增量修改

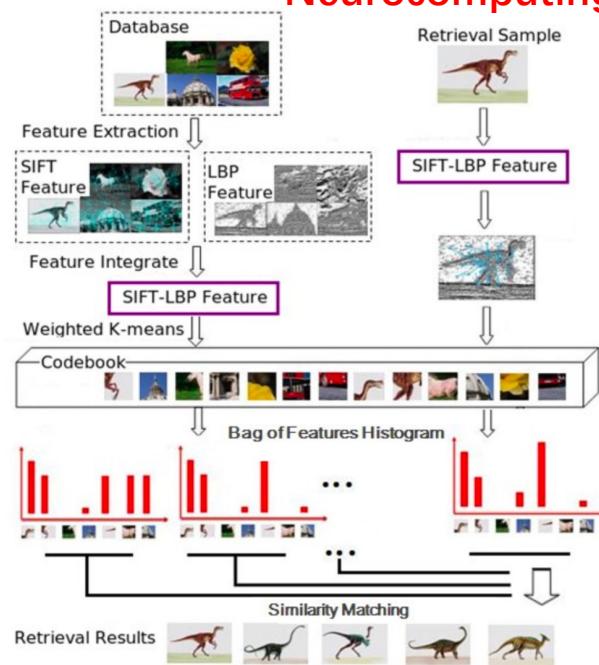
Feature integration analysis of bag-of-features model for image retrieval

Jing Yu^a, Zengchang Qin^{a,*}, Tao Wan^b, Xi Zhang^a

^aIntelligent Computing and Machine Learning Lab, School of Automation Science and Electrical Engineering, Beihang University, Beijing, China

^bSchool of Medicine, Boston University, Boston, USA

Neurocomputing 2011



从科学问题是否探究本质说起

CCF—A

- 问题-方法-实验，相互呼应

- 问题：有理有据，足够具体
- 方法：针对问题设计，每一步设计目标明确
- 实验：针对方法逐一证明，针对动机逐一分析

CCF—C

- 问题-方法-实验，各为其说

- 问题：大家都在研究，所以我研究
- 方法：step1->step2->step3
- 实验：达到了SOTA，缺乏分析

一篇论文的组成

- ◆ **标题：**核心问题与创新点的高度凝练
- ◆ **摘要：**动机、亮点与效果的概要介绍
- ◆ **引言：**动机、现状、方法和贡献介绍
- ◆ **相关工作：**引言现有工作的详细介绍
- ◆ **研究方法：**引言模型部分的详细介绍
- ◆ **实验分析：**引言模型效果的详细介绍
- ◆ **总结展望：**经过实验验证后给的结论
- ◆ **致谢：**对作者以外人员/机构的感谢
- ◆ **参考文献：**正文出现的按照格式引用



一篇论文的组成——标题

基本要求（不超过15个单词）

☀ 英文形式规范

☀ 语言精炼简洁

☀ 范围大小适当

好标题

☀ 反应核心问题

☀ 突出技术创新

☀ 保护知识产权

☀ 易于记忆传播

一篇论文的组成——标题 (Bad Cases)

From shallow to deeper: compositional reasoning
over graphs for visual question answering

太宽泛!

PERT: **adaPtive** Evidence-driven Reasoning **neTwork** for Machine Reading
Comprehension with Unanswerable Questions

不规范!

Understanding like humans: multimodal representation for the visual
information in visual dialog

没依据!

Graph Neural Networks for Image-Text Matching

没创新!

A **Plug-and-Play novel Tree Loss Function** for Unbiased Scene Graph
Generation based on **Upgraded Transformer framework**

太冗余!

KBGN: Knowledge-Bridge Graph Network for Adaptive Vision-Text Reasoning
in Visual Dialogue.

难记忆!

一篇论文的组成——标题 (My Own)

MuKEA: Multimodal Knowledge Extraction and Accumulation for Knowledge-based Visual Question Answering (CVPR 2022)

ET-BERT: A Contextualized Datagram Representation with Pre-training Transformers for Encrypted Traffic Classification (WWW 2022)

CogTree: Cognition Tree Loss for Unbiased Scene Graph Generation (IJCAI 2021)

DualDV: An Adaptive Dual Encoding Model for Visual Dialogue (AAAI 2020)

Mucko: Multi-Layer Cross-Modal Knowledge Reasoning for Fact-based Visual Question Answering (IJCAI 2020)

DAM: Deliberation, Abandon and Memory Networks for Generating Detailed and Non-repetitive Responses in Visual Dialogue (IJCAI 2020)

一篇论文的组成——标题 (CV & NLP & ML)

Zero-Shot Text-to-Image Generation
(DALL·E, arxiv 2021)

Swin Transformer: Hierarchical Vision Transformer using Shifted Windows (ICCV 2021)

Sketch, Ground, and Refine: Top-Down Dense Video Captioning (CVPR 2021)

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding
(NAACL 2019)

Knowledgeable Prompt-tuning : Incorporating Knowledge into Prompt Verbalizer for
Text Classification (ACL 2022)

Semi-Supervised Classification with Graph Convolutional Networks (ICLR 2017)

A Simple Framework for Contrastive Learning of Visual Representations
(SimCLR, 2020 ICML)

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一篇论文的组成——摘要

基本要求

☀ 标题的扩充、引言的概括

☀ 涵盖动机、亮点、效果等

☀ 200词左右

☀ 逻辑清晰

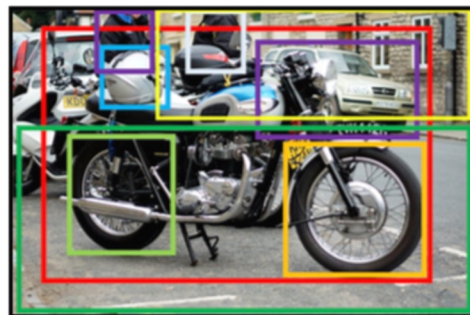
一篇论文的组成——摘要

CVPR 2020

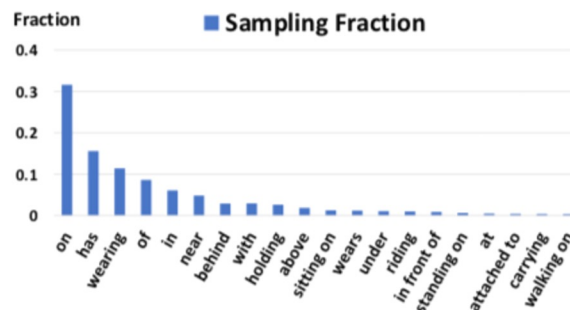
Unbiased Scene Graph Generation from Biased Training

Kaihua Tang¹, Yulei Niu³, Jianqiang Huang^{1,2}, Jiaxin Shi⁴, Hanwang Zhang¹

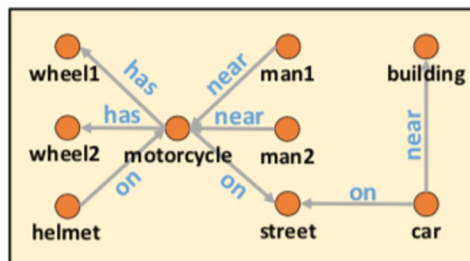
¹Nanyang Technological University, ²Damo Academy, Alibaba Group, ³Renmin University of China, ⁴Tsinghua University



(a) Input Image



(b) Distribution of Predicate Sampling Fraction



(c) Biased Generation



(d) Unbiased Generation



一篇论文的组成—摘要

Unbiased Scene Graph Generation from Biased Training CVPR 2020

Today's scene graph generation (SGG) task is still far from practical, mainly due to the severe training bias, e.g., collapsing diverse human walk on/ sit on/lay on beach into human on beach. Given such SGG, the down-stream tasks such as VQA can hardly infer better scene structures than merely a bag of objects. However, debiasing in SGG is not trivial because traditional debiasing methods cannot distinguish between the good and bad bias, e.g., good context prior (e.g., person read book rather than eat) and bad long-tailed bias (e.g., near dominating behind/in front of). In this paper, we present a novel SGG framework based on **causal inference** but not the conventional likelihood. We first build a causal graph for SGG, and perform traditional biased training with the graph. Then, we propose to draw the **counterfactual causality** from the trained graph to infer the effect from the bad bias, which should be removed. In particular, we use **Total Direct Effect** as the proposed final predicate score for unbiased SGG. Note that our framework is agnostic to any SGG model and thus can be widely applied in the community who seeks unbiased predictions. By using the proposed **Scene Graph Diagnosis** toolkit¹ on the SGG benchmark Visual Genome and several prevailing models, we observed significant improvements over the previous state-of-the-art methods.

SGG存在的挑战：数据偏置

SGG方法的问题：未区分偏置的好坏

本文方法的思路：因果推断

本文具体的亮点：

构建因果图-->反事实推断-->优化目标

本文方法的优势：

- 模型无关，广泛适用
- 模型效果达到新SOTA



一篇论文的组成——引言

基本要求（摘要的扩展，最难的部分**）**

☀ 研究背景与挑战

☀ 提出问题与原因

☀ 相关工作与不足

☀ 本文研究思路

☀ 本文主要贡献



一篇论文的组成——引言

CCF—A

- 问题-方法-实验，相互呼应

- 问题：有理有据，足够具体

- 背景阐述聚焦重点

- 问题提出明确具体

- 聚焦研究动机，总结现状问题

- 基于研究动机，概述研究方法

- 面向领域需求，拔高论文贡献

- 方法：针对问题设计，每一步设计目标明确

- 实验：针对方法逐一证明，针对动机逐一分析

CCF—C

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一篇论文的组成——引言

IJCAI 2020

Mucko: Multi-Layer Cross-Modal Knowledge Reasoning for Fact-based Visual Question Answering

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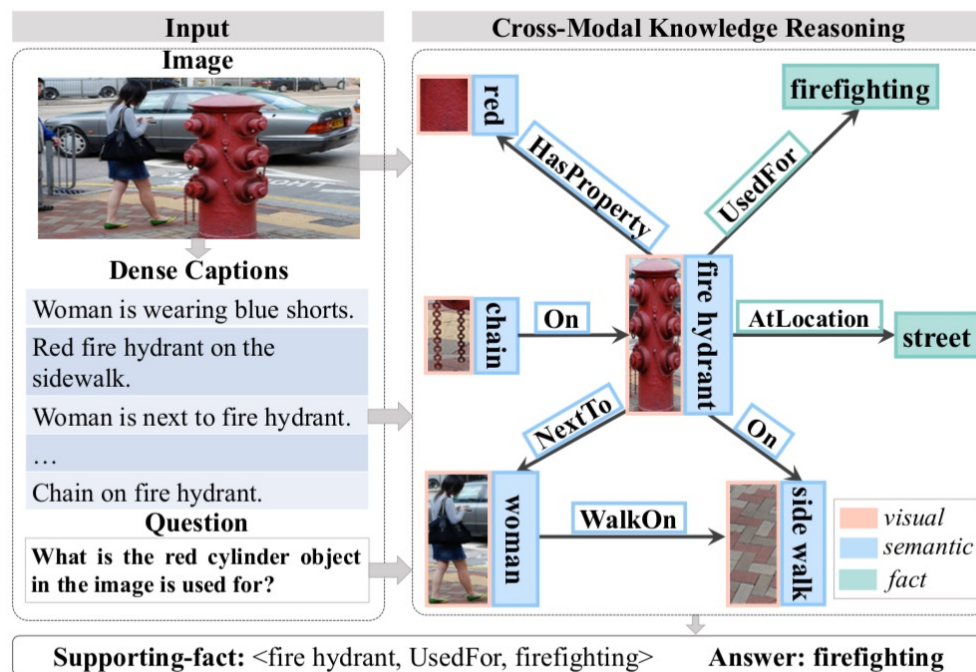
¹Institute of Information Engineering, Chinese Academy of Sciences, Beijing, China

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³Microsoft Research Asia, Beijing, China

⁴University of Adelaide, Australia

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一篇论文的组成——引言

Abstract

Fact-based Visual Question Answering (FVQA) requires external knowledge beyond visible content to answer questions about an image, which is challenging but indispensable to achieve general VQA. One limitation of existing FVQA solutions is that they jointly embed all kinds of information without fine-grained selection, which introduces unexpected noises for reasoning the final answer. How to capture the question-oriented and information-complementary evidence remains a key challenge to solve the problem. In this paper, we depict an image by a multi-modal heterogeneous graph, which contains multiple layers of information corresponding to the visual, semantic and factual features. On top of the multi-layer graph representations, we propose a modality-aware heterogeneous graph convolutional network to capture evidence from different layers that is most relevant to the given question. Specifically, the intra-modal graph convolution selects evidence from each modality and cross-modal graph convolution aggregates relevant information across different modalities. By stacking this process multiple times, our model performs iterative reasoning and predicts the optimal answer by analyzing all question-oriented evidence. We achieve a new state-of-the-art performance on the FVQA task and demonstrate the effectiveness and interpretability of our model with extensive experiments. The code is available at <https://github.com/astro-zihao/mucko>.

1 Introduction

Visual question answering (VQA) [Antol *et al.*, 2015] is an attractive research direction aiming to jointly analyze multi-modal content from images and natural language. Equipped with the capacities of grounding, reasoning and translating, a VQA agent is expected to answer a question in natural language based on an image. Recent works [Cadene *et al.*, 2019;

*Equal contribution.

†Corresponding author.

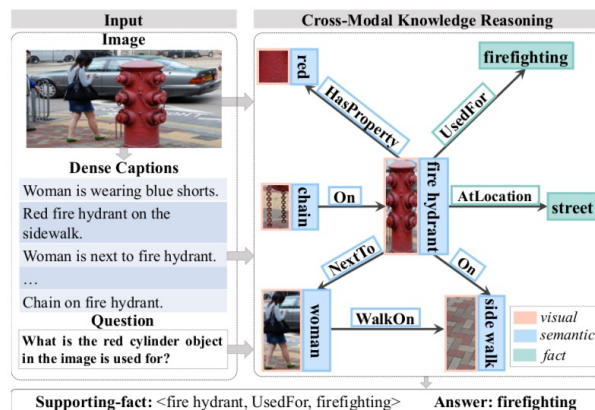


Figure 1: An illustration of our motivation. We represent an image by multi-layer graphs and cross-modal knowledge reasoning is conducted on the graphs to infer the optimal answer.

Li *et al.*, 2019b; Ben-Younes *et al.*, 2019] have achieved great success in the VQA problems that are answerable by solely referring to the visible content of the image. However, such kinds of models are incapable of answering questions which require external knowledge beyond what is in the image. Considering the question in Figure 1, the agent not only needs to visually localize ‘the red cylinder’, but also to semantically recognize it as ‘fire hydrant’ and connects the knowledge that ‘fire hydrant is used for firefighting’. Therefore, how to collect the question-oriented and information-complementary evidence from visual, semantic and knowledge perspectives is essential to achieve general VQA.

To advocate research in this direction, [Wang *et al.*, 2018] introduces the ‘Fact-based’ VQA (FVQA) task for answering questions by joint analysis of the image and the knowledge base of facts. The typical solutions for FVQA build a fact graph with fact triplets filtered by the visual concepts in the image and select one entity in the graph as the answer. Existing works [Wang *et al.*, 2017; Wang *et al.*, 2018] parse the question as keywords and retrieve the supporting-entity only by keyword matching. This kind of approaches is vulnerable when the question does not exactly mention the visual concepts (*e.g.* synonyms and homographs) or the mentioned information is not captured in the fact graph (*e.g.* the visual

attribute ‘red’ in Figure 1 may be falsely omitted). To resolve these problems, [Narasimhan *et al.*, 2018] introduces visual information into the fact graph and infers the answer by implicit graph reasoning under the guidance of the question. However, they provide the whole visual information equally to each graph node by concatenation of the image, question and entity embeddings. Actually, only part of the visual content are relevant to the question and a certain entity. Moreover, the fact graph here is still homogeneous since each node is represented by a fixed form of image-question-entity embedding, which limits the model’s flexibility of adaptively capturing evidence from different modalities.

In this work, we depict an image as a multi-modal heterogeneous graph, which contains multiple layers of information corresponding to different modalities. The proposed model is focused on **Multi-Layer Cross-Modal Knowledge Reasoning** and we name it as **Mucko** for short. Specifically, we encode an image by three layers of graphs, where the object appearance and their relationships are kept in the *visual layer*, the high-level abstraction for bridging the gaps between visual and factual information is provided in the *semantic layer*, and the corresponding knowledge of facts are supported in the *fact layer*. We propose a modality-aware heterogeneous graph convolutional network to adaptively collect complementary evidence in the multi-layer graphs. It can be performed by two procedures. First, the Intra-Modal Knowledge Selection procedure collects question-oriented information from each graph layer under the guidance of question; Then, the Cross-Modal Knowledge Reasoning procedure captures complementary evidence across different layers.

The main contributions of this paper are summarized as follows: (1) We comprehensively depict an image by a heterogeneous graph containing multiple layers of information based on visual, semantic and knowledge modalities. We consider these three modalities jointly and achieve significant improvement over state-of-the-art solutions. (2) We propose a modality-aware heterogeneous graph convolutional network to capture question-oriented evidence from different modalities. Especially, we leverage an attention operation in each convolution layer to select the most relevant evidence for the given question, and the convolution operation is responsible for adaptive feature aggregation. (3) We demonstrate good interpretability of our approach and provide case study in deep insights. Our model automatically tells which modality (visual, semantic or factual) and entity have more contributions to answer the question through visualization of attention weights and gate values.

一篇论文的组成——引言

Abstract

Fact-based Visual Question Answering (FVQA) requires external knowledge beyond visible content to answer questions about an image, which is challenging but indispensable to achieve general VQA. One limitation of existing FVQA solutions is that they jointly embed all kinds of information without fine-grained selection, which introduces unexpected noises for reasoning the final answer. How to capture the question-oriented and information-complementary evidence remains a key challenge to solve the problem. In this paper, we depict an image by a multi-modal heterogeneous graph, which contains multiple layers of information corresponding to the visual, semantic and factual features. On top of the multi-layer graph representations, we propose a modality-aware heterogeneous graph convolutional network to capture evidence from different layers that is most relevant to the given question. Specifically, the intra-modal graph convolution selects evidence from each modality and cross-modal graph convolution aggregates relevant information across different modalities. By stacking this process multiple times, our model performs iterative reasoning and predicts the optimal answer by analyzing all question-oriented evidence. We achieve a new state-of-the-art performance on the FVQA task and demonstrate the effectiveness and interpretability of our model with extensive experiments. The code is available at <https://github.com/astro-zihao/mucko>.

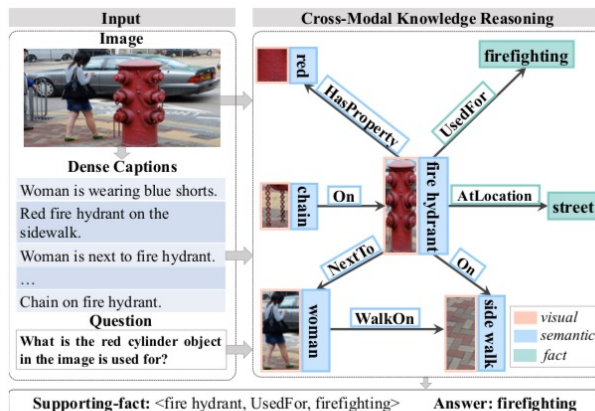


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The main contributions of this paper are as follows: (1) We comprehensively depict an image by a heterogeneous graph containing multiple layers of information based on visual, semantic and knowledge modalities. We consider these three modalities jointly and achieve significant improvement over state-of-the-art solutions. (2) We propose a modality-aware heterogeneous graph convolutional network to capture question-oriented evidence from different modalities. Especially, we leverage an attention operation in each convolution layer to select the most relevant evidence for the given question, and the convolution operation is responsible for adaptive feature aggregation. (3) We demonstrate good interpretability of our approach and provide case study in deep insights. Our model automatically tells which modality (visual, semantic or factual) and entity have more contributions to answer the question through visualization of attention weights and gate values.

从具体方法归纳问题

可解决的小问题

方法介绍：呼应挑战和问题

如何解决问题

贡献：凝练方法的普适性

提出的新问题？
解决的新视角？
实现的新框架？
实现的新方法？
达到的新SOTA？
具备的新能力？

有主要实验结果后可以动笔！

1 Introduction

大背景

Visual question answering (VQA) [Antol et al., 2015] is an attractive research direction aiming to jointly analyze multi-modal content from images and natural language. Equipped with the capacities of grounding, reasoning and translating, a VQA agent is expected to answer a question in natural language based on an image. Recent works [Cadene et al., 2019;

背景：VQA需要知识
挑战：引入互补知识

现有方法问题：分类归纳

准确、专业评价

程研究所

CogModal



一篇论文的组成——相关工作

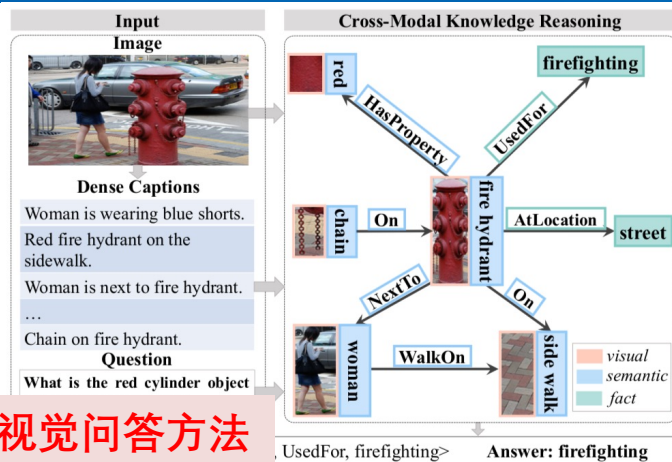
基本要求

- ☀ 包括理解本文的所有主题
- ☀ 包括问题相关的所有工作
- ☀ 从不同维度划分主题
- ☀ 同一主题方法归类
- ☀ 总结问题
- ☀ 引出本研究的区别和贡献

不要所有看过的论文！
不要罗列写上的论文！



一篇论文的组成——相关工作



视觉问答方法

Visual Question Answering. The typical solutions for VQA are based on the CNN-RNN architecture [Malinowski *et al.*, 2015] and leverage global visual features to represent image, which may introduce noisy information. Various attention mechanisms [Yang *et al.*, 2016; Lu *et al.*, 2016; Anderson *et al.*, 2018] have been exploited to highlight visual objects that are relevant to the question. However, they treat objects independently and ignore their informative relationships. [Battaglia *et al.*, 2018] demonstrates that human's ability of combinatorial generalization highly depends on the mechanisms for reasoning over relationships. Consistent with such proposal, there is an emerging trend to represent the image by graph structure to depict objects and relationships in VQA and other vision-language tasks [Hu *et al.*, 2019b; Wang *et al.*, 2019a; Li *et al.*, 2019b]. As an extension, [Jiang *et al.*, 2020] exploits natural language to enrich the graph-based visual representations. However, it solely captures the semantics in natural language by LSTM, which lacking of fine-grained correlations with the visual information. To go one step further, we depict an image by multiple layers of graphs from visual, semantic and factual perspectives to collect fine-grained evidence from different modalities.

基于知识的视觉问答方法

Fact-based Visual Question Answering. Human can easily combine visual observation with external knowledge for answering questions, which remains challenging for algorithms. [Wang *et al.*, 2018] introduces a fact-based VQA task, which provides a knowledge base of facts and associates each question with a supporting-fact. Recent works based on FVQA generally select one entity from fact graph as the answer and falls into two categories: query-mapping based methods and learning based methods. [Wang *et al.*, 2017] reduces the question to one of the available query templates and this limits the types of questions that can be asked. [Wang *et al.*, 2018] automatically classifies and maps the question to a query which does not suffer the above constraint. Among both methods, however, visual information are used to extract facts but not introduced during the reasoning process. [Narasimhan *et al.*, 2018] applies GCN on the fact graph where each node is represented by the fixed form of image-question-entity embedding. However, the visual information is wholly provided which may introduce redundant information for prediction. In this paper, we depict an image by multi-layer graphs and perform cross-modal heterogeneous graph reasoning on them to capture complementary evidence from different layers that most relevant to the question.

Heterogeneous Graph Neural Networks. Graph neural network in the last few years has gained momentum in the last few years and with homogeneous graphs, heterogeneous graphs are more common in the real world. [Schlichtkrull *et al.*, 2018] generalizes graph convolutional network (GCN) to handle different relationships between entities in a knowledge base, where each edge with distinct relationships is encoded independently. [Wang *et al.*, 2019b; Hu *et al.*, 2019a] propose heterogeneous graph attention networks with dual-level attention mechanism. All of these methods model different types of nodes and edges on a unified graph. In contrast, the heterogeneous graph in this work contains multiple layers of subgraphs and each layer consists of nodes and edges coming from different modalities. For this specific constrain, we propose the intra-modal and cross-modal graph convolutions for reasoning over such multi-modal heterogeneous graphs.

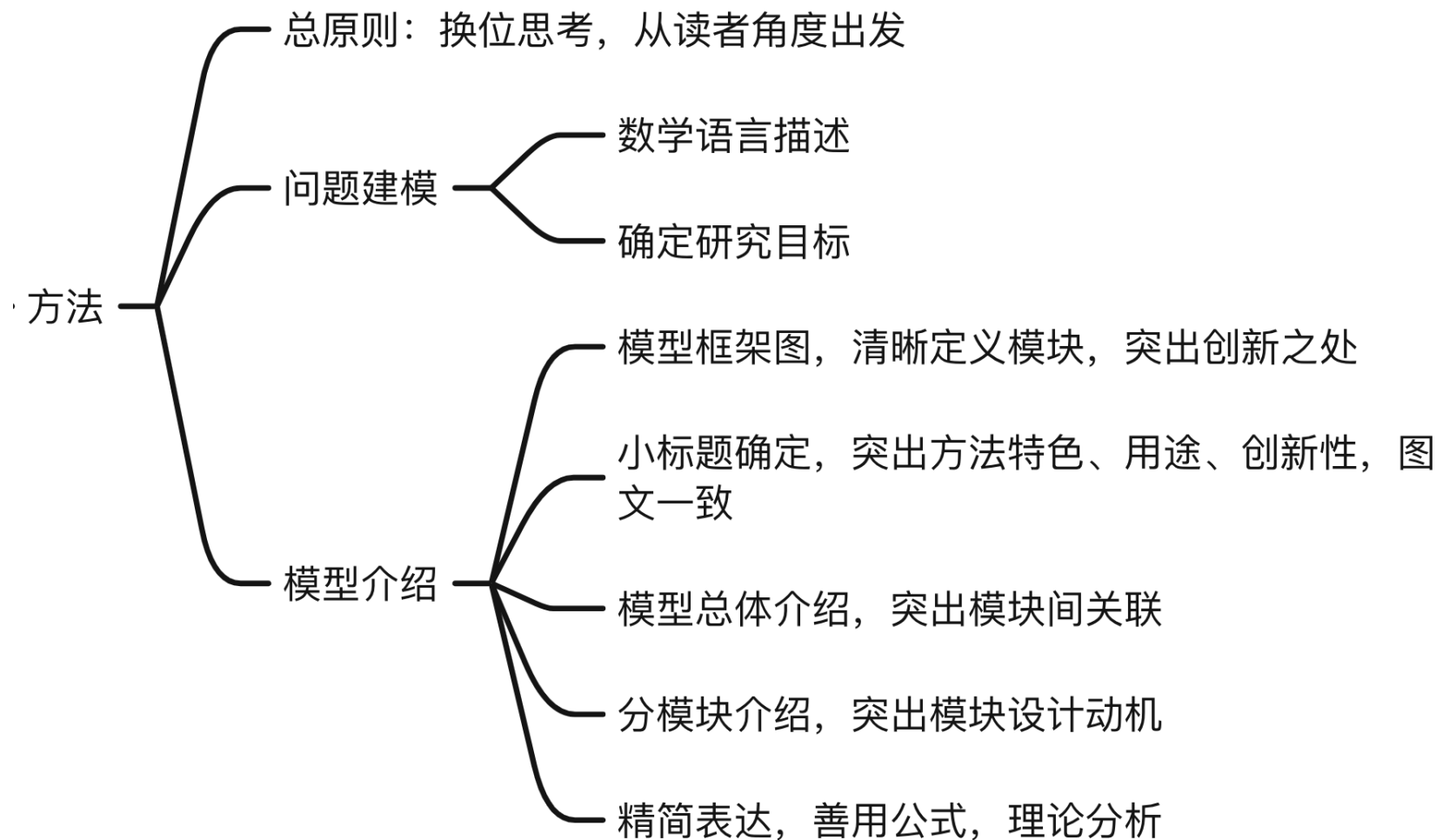
异构图神经网络方法

方法写完后可动笔!



一篇论文的组成——方法

基本要求（最容易部分，可以先写）



一篇论文的组成——方法

CCF—A

- 问题-方法-实验，相互呼应
 - 动机：有理有据，足够具体
 - 方法：针对问题设计，每一步设计目标明确
 - 根据重点，重新组织方法介绍思路
 - 标题和图突出创新性和重点，相互呼应
 - 每一步方法设计都有理可依
 - 实验：针对方法逐一证明，针对动机逐一分析

CCF—C

- 问题-方法-实验，各为其说
 - 动机：大家都在研究，所以我研究
 - 方法：step1->step2->step3
 - 实验：达到了SOTA，缺乏分析

一篇论文的组成——方法

IJCAI 2020

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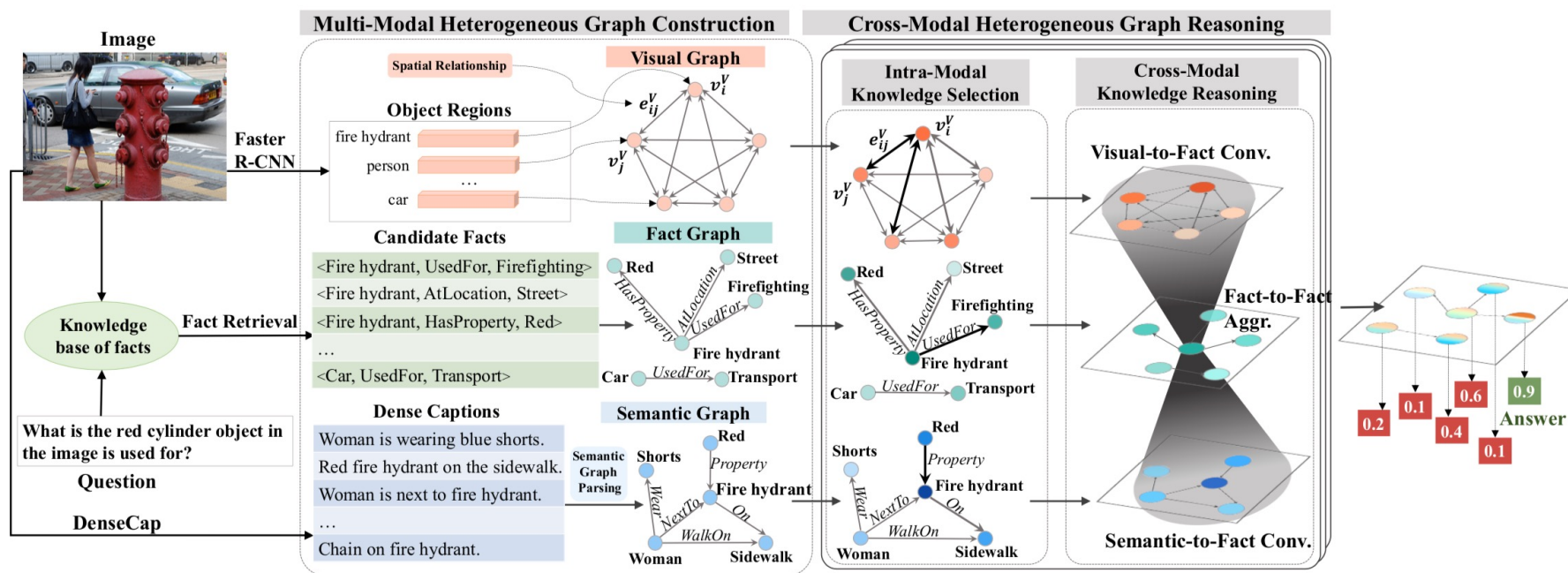
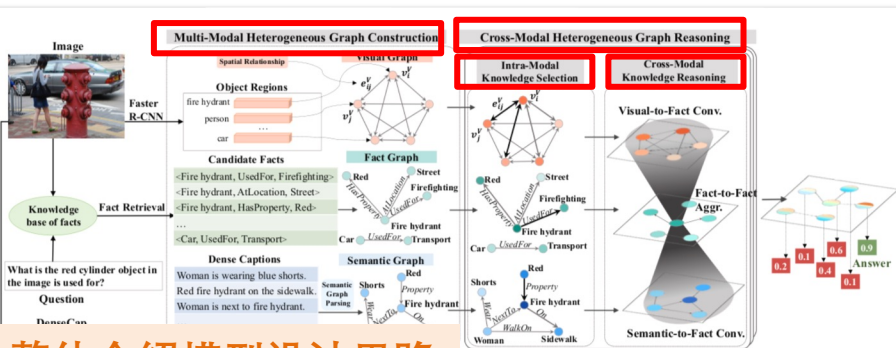


Figure 2: An overview of our model. The model contains two modules: Multi-modal Heterogeneous Graph Construction aims to depict an image by multiple layers of graphs and Cross-modal Heterogeneous Graph Reasoning supports intra-modal and cross-modal evidence selection.



一篇论文的组成——方法



整体介绍模型设计思路

where e_1 is a visual concept in the image, e_2 is an attribute or phrase and r represents the relationship between e_1 and e_2 . The key is to choose a correct entity, i.e. either e_1 or e_2 , from the supporting fact as the predicted answer. We first introduce a novel scheme of depicting an image by three layers of graphs, including the visual graph, semantic graph and fact graph respectively, imitating the understanding of various properties of an object and the relationships. Then we perform cross-modal heterogeneous graph reasoning that consists of two parts: *Intra-Modal Knowledge Selection* aims to choose question-oriented knowledge from each layer of graphs by intra-modal graph convolutions, and *Cross-Modal Knowledge Reasoning* adaptively selects complementary evidence across three layers of graphs by cross-modal graph convolutions. By stacking the above two processes multiple times, our model performs iterative reasoning across all the modalities and results in the optimal answer by jointly analyzing all the entities. Figure 2 gives detailed illustration of our model.

3.1 Multi-Modal Graph Construction

Visual Graph Construction. Since most of the questions in FVQA grounded in the visual objects and their relationships, we construct a fully-connected visual graph to present such evidence at appearance level. Given an image, we use Faster-RCNN [Ren et al., 2017] to identify a set of objects $\mathcal{O} = \{o_i\}_{i=1}^K$ ($K = 36$), where each object o_i is associated with a visual feature vector $v_i \in \mathbb{R}^{d_v}$ ($d_v = 2048$), a spatial feature vector $b_i \in \mathbb{R}^{d_b}$ ($d_b = 4$) and a corresponding label. Specifically, $b_i = [x_i, y_i, w_i, h_i]$ where (x_i, y_i) is the bounding box of o_i and w_i, h_i are the width and height of o_i .

每一个过程首先介绍背后的动机和目标

node set and each node v_i^V corresponds to a detected object o_i . The feature of node v_i^V is represented by v_i^V . Each edge $e_{ij}^V \in \mathcal{E}^V$ denotes the relative spatial relationships between two objects. We encode the edge feature by a 5-dimensional vector, i.e. $r_{ij}^V = [\frac{x_j - x_i}{w_i}, \frac{y_j - y_i}{h_i}, \frac{w_j}{w_i}, \frac{h_j}{h_i}, \frac{w_j h_j}{w_i h_i}]$.

Multi-modal Heterogeneous Graph Construction aims to depict an image by three layers of graphs, including the visual graph, semantic graph and fact graph respectively, imitating the understanding of various properties of an object and the relationships. Then we perform cross-modal heterogeneous graph reasoning that consists of two parts: *Intra-Modal Knowledge Selection* aims to choose question-oriented knowledge from each layer of graphs by intra-modal graph convolutions, and *Cross-Modal Knowledge Reasoning* adaptively selects complementary evidence across three layers of graphs by cross-modal graph convolutions. By stacking the above two processes multiple times, our model performs iterative reasoning across all the modalities and results in the optimal answer by jointly analyzing all the entities. Figure 2 gives detailed illustration of our model.

Semantic Graph Construction. In addition to visual information, high-level abstraction of the objects and relationships by natural language provides essential semantic information. Such abstraction is indispensable to associate the visual objects in the image with the concepts mentioned in both questions and facts. In our work, we leverage dense captions [Johnson et al., 2016] to extract a set of local-level semantics in an image, ranging from the properties of a single object (color, shape, emotion, etc.) to the relationships between objects (action, spatial positions, comparison, etc.). We depict an image by D dense captions, denoted as $Z = \{z_i\}_{i=1}^D$, where z_i is a natural language description about a local region in the image. Instead of using monolithic embeddings to represent the captions, we exploit to model them by a graph-based semantic representation, denoted as $\mathcal{G}^S = (\mathcal{V}^S, \mathcal{E}^S)$, which is constructed by a semantic graph parsing model [Anderson et al., 2016]. The node $v_i^S \in \mathcal{V}^S$ represents the name or attribute of an object extracted from the captions while the edge $e_{ij}^S \in \mathcal{E}^S$ represents the relationship between v_i^S and v_j^S .

- 标题突出创新点和过程
- 表达逻辑一致
- 图文一致

base of facts following a scored based approach proposed in [Narasimhan et al., 2018]. We compute the cosine similarity of the embeddings of every word in the fact with the words in the question.

cepts detected in the question. We assign a similarity score to each concept based on the similarity between the concept and the words in the question. The concepts with the highest similarity are retained, denoted as f_{100} . A relation type classifier is trained additionally to further filter the retrieved facts. Specifically, we feed the last hidden state of LSTM to an MLP layer to predict the relation type $\hat{r}_i$ of a question. We retain the facts among f_{100} only if their relationships agree with $\hat{r}_i$, i.e. $f_{rel} = f$ if $f_{rel} : r(f) \in \{\hat{r}_i\}$ ($\{\hat{r}_i\}$ contains top-3 predicted relationships in experiments). Then a fact graph $\mathcal{G}^F = (\mathcal{V}^F, \mathcal{E}^F)$ is built upon f_{rel} as the candidate facts can be naturally organized as graphical structure. Each node $v_i^F \in \mathcal{V}^F$ denotes an entity in f_{rel} and is represented by GloVe embedding of the entity, denoted as v_i^F . Each edge $e_{ij}^F \in \mathcal{E}^F$ denotes the relationship between v_i^F and v_j^F and is represented by GloVe embedding r_{ij} . The topological structure among facts can be effectively exploited by jointly considering all the entities in the fact graph.

3.2 Intra-Modal Knowledge Selection

Since each layer of graphs contains modality-specific knowledge relevant to the question, we first select valuable evidence independently from the visual graph, semantic graph and fact graph by **Visual-to-Visual Convolution**, **Semantic-to-Semantic Convolution** and **Fact-to-Fact Convolution** respectively. These three convolutions share the common operations but differ in their node and edge representations corresponding to the graph layers. Thus we omit the superscript of node representation v and edge representation r in the rest of this section. We first perform attention operations to highlight the nodes and edges that are most relevant to the question q and consequently update node representations via intra-modal graph convolution. This process mainly consists of the following three steps:

Question-guided Node Attention. We first evaluate the relevance of each node corresponding to the question by attention mechanism. The attention weight for v_i is computed as:

$$\alpha_i = \text{softmax}(w_a^T \tanh(W_1 v_i + W_2 q)) \quad (1)$$

where W_1, W_2 and w_a (as well as $W_3, \dots, W_6, w_b, w_c$ mentioned below) are learned parameters. q is question embedding encoded by LSTM.

Question-guided Edge Attention. Under the guidance of question, we then evaluate the importance of edge e_{ji} constrained by the neighbor node v_j regarding to v_i as:

$$\beta_{ji} = \text{softmax}(w_b^T \tanh(W_3 v_j' + W_4 q')) \quad (2)$$

where $v_j' = W_5[v_j, r_{ji}]$, $q' = W_6[v_i, q]$ and $[\cdot, \cdot]$ denotes concatenation operation.

Intra-Modal Graph Convolution. Given the node and edge attention weights learned in Eq. 1 and Eq. 2, the node representations of each layer of graphs are updated following

具体过程分小标题

$$m_i = \sum_{j \in \mathcal{N}_i} \beta_{ji} v_j' \quad (3)$$

$$\tilde{v}_i = \text{ReLU}(W_7[m_i, \alpha_i v_i]) \quad (4)$$

where $\mathcal{N}_i$ is the neighborhood set of node v_i .

We conduct the above intra-modal knowledge selection on $\mathcal{G}^V$, $\mathcal{G}^S$ and $\mathcal{G}^F$ independently and obtain the updated node representations, denoted as $\{\tilde{v}_i^V\}_{i=1}^{N^V}$, $\{\tilde{v}_i^S\}_{i=1}^{N^S}$ and $\{\tilde{v}_i^F\}_{i=1}^{N^F}$ accordingly.

3.3 Cross-Modal Knowledge Reasoning

To answer the question correctly, we fully consider the complementary evidence from visual, semantic and factual information. Since the answer comes from one entity in the fact graph, we gather complementary information from visual graph and semantic graph to fact graph by cross-modal convolutions, including **visual-to-fact convolution** and **semantic-to-fact convolution**. Finally, a **fact-to-fact aggregation** is performed on the fact graph to reason over all the entities and form a global decision.

Visual-to-Fact Convolution. For the entity v_i^F in fact graph, the attention value of each node v_j^V in the visual graph w.r.t. v_i^F is calculated under the guidance of question:

$$\gamma_{ji}^{V \rightarrow F} = \text{softmax}(w_c \tanh(W_8 v_j^V + W_9 [v_i^F, q])) \quad (5)$$

The complementary information $m_i^{V \rightarrow F}$ from visual graph for v_i^F is computed as:

$$m_i^{V \rightarrow F} = \sum_{j \in \mathcal{N}^V} \gamma_{ji}^{V \rightarrow F} v_j^V \quad (6)$$

Semantic-to-Fact Convolution. The complementary information $m_i^{S \rightarrow F}$ from the semantic graph is computed in the same way as in Eq. 5 and Eq. 6.

Then we fuse the complementary knowledge for v_i^F from three layers of graphs via a gate operation:

$$\text{gate}_i = \sigma(W_{10}[m_i^{V \rightarrow F}, m_i^{S \rightarrow F}, \tilde{v}_i^F]) \quad (7)$$

$$\tilde{v}_i^F = W_{11}[\text{gate}_i \circ [m_i^{V \rightarrow F}, m_i^{S \rightarrow F}, \tilde{v}_i^F]] \quad (8)$$

where σ is sigmoid function and " $\circ$ " denotes element-wise product.

Fact-to-Fact Aggregation. Given a set of candidate entities in the fact graph $\mathcal{G}^F$, we aim to globally compare all the entities and select an optimal one as the answer. Now the representation of each entity in the fact graph gathers question-oriented information from three modalities. To jointly evaluate the possibility of each entity, we perform the attention-based graph convolutional network similar to Fact-to-Fact Convolution introduced in Section 3.2 to aggregate information in the fact graph and obtain the transformed entity representations.

We iteratively perform intra-modal knowledge selection and cross-modal knowledge reasoning in multiple steps to obtain the final entity representations. After T steps, each entity representation $\tilde{v}_i^{F(T)}$ captures the structural information within T -hop neighborhood across three layers.

3.4 Learning

The concatenation of entity representation $\tilde{v}_i^{F(T)}$ and question embedding q is passed to a binary classifier to predict its probability as the answer, i.e. $\hat{y}_i = p_\theta([\tilde{v}_i^{F(T)}, q])$. We apply the binary cross-entropy loss in the training process:

$$l_i = - \sum_{i \in \mathcal{N}^F} [a \cdot y_i \ln \hat{y}_i + b \cdot (1 - y_i) \ln (1 - \hat{y}_i)] \quad (9)$$

where y_i is the ground truth label for $v_i^{F(T)}$ and a, b represent loss function weights for positive and negative samples respectively. The entity with the largest probability is selected as the final answer.

注意：
动机？
其他方式？

一篇论文的组成——实验

基本要求

☀️ **一致**：支撑理论/方法、动机

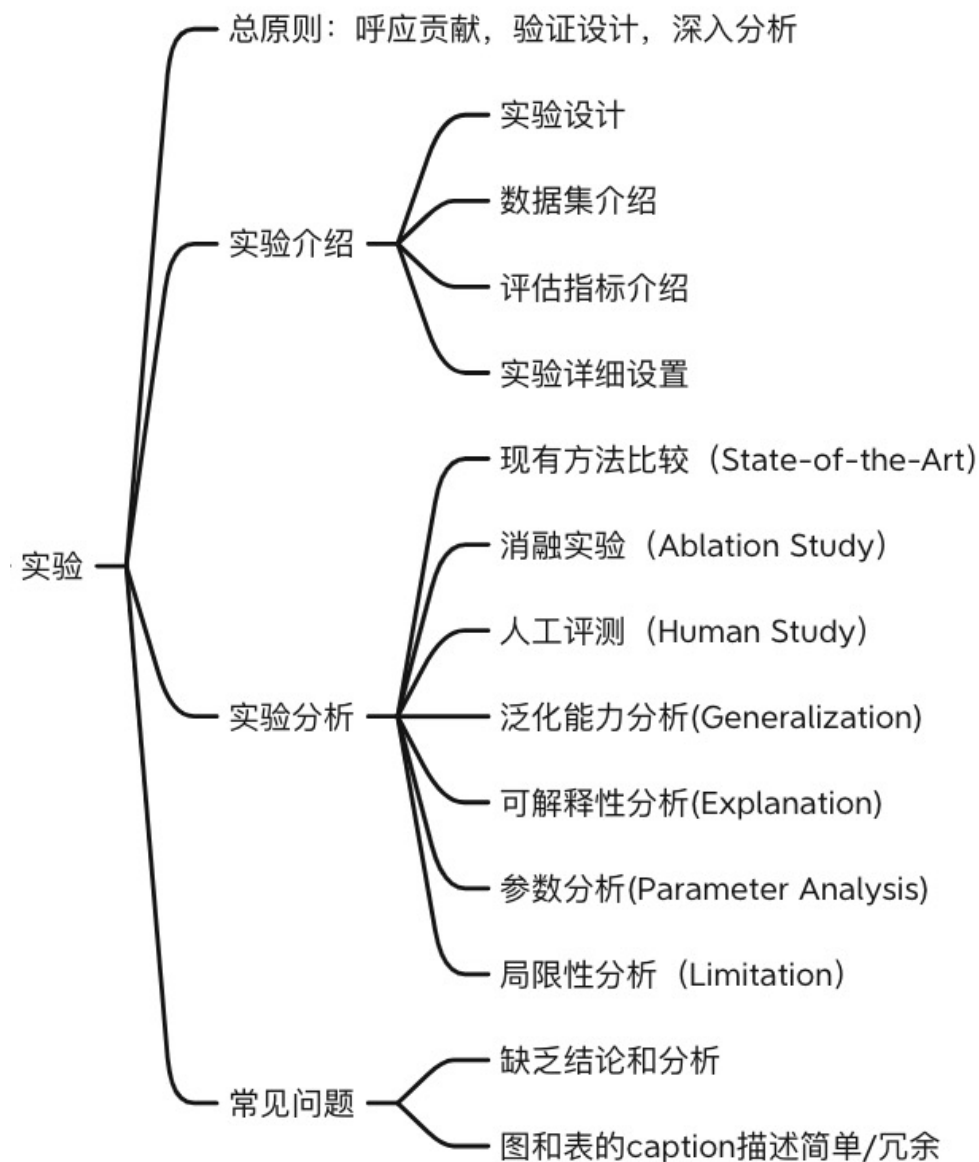
☀️ **核心**：提供重要实验结果

☀️ **诚实**：不只展示最佳个例

☀️ **分析**：给出结果的合理解释

☀️ **局限**：给出方法的能力边界

学习优秀论文的实验写法！



一篇论文的组成——实验

CCF—A

- 问题-方法-实验，相互呼应

- 问题：有理有据，足够具体
- 方法：针对问题设计，每一步设计目标明确
- 实验：针对方法逐一证明，针对动机逐一分析

- 结论先行
- 事实支撑
- 层次分明

CCF—C

- 问题-方法-实验，各为其说

- 问题：大家都在研究，所以我研究
- 方法：step1->step2->step3
- 实验：达到了SOTA，缺乏分析



一篇论文的组成——实验

Table 1

State-of-the-art comparison on FVQA dataset.

Method	Overall accuracy	
	top-1	top-3
LSTM-Q + Image + Pre-VQA [4]	24.98	40.40
Hie-Q + Image + Pre-VQA [4]	43.14	59.44
FVQA (top-3-QQmapping) [4]	56.91	64.65
FVQA (Ensemble) [4]	58.76	-
Straight to the Facts (STTF) [19]	62.20	75.60
Reading Comprehension [43]	62.96	70.08
Out of the Box (OB) [6]	69.35	80.25
Human [4]	77.99	-
GRUC (ours)	69.03	88.12

对比方法分类介绍

Table 3 State-of-the-art comparison on OK-VQA dataset.

We also report the quantitative performance on the challenging OK-VQA dataset in Table 3. We compare our model with three kinds of existing models, including current state-of-the-art VQA models, knowledge-based VQA models and ensemble models. The VQA models contain Q-Only [21], MLP [21], BAN [44], MUTAN [44]. The knowledge-based VQA models [21] consist of ArticleNet (AN), BAN + AN and MUTAN + AN. The ensemble models [21], i.e. BAN/AN oracle and MUTAN/AN oracle, simply take the raw ArticleNet and VQA model predictions, taking the best answer (comparing to ground truth) from either. We report the overall performance (top-1 and top-3 accuracy) as well as breakdowns for each of the knowledge categories (top-1 accuracy). We have the following two observations from the results:

现有方法分类对比结果分析

As all the compared state-of-the-art models (BAN and MUTAN) specifically designed for VQA tasks, they get inferior results compared with ours. This indicates that general VQA task like OK-VQA cannot be simply solved by a well-designed model, but requires the ability to incorporate external knowledge in an effective way. Moreover, our model outperforms knowledge-based VQA models including both single models (BAN+AN and MUTAN + AN) and ensemble models (BAN/AN oracle and MUTAN/AN oracle), which further proves the advantages of our knowledge incorporating mechanism based on both multimodal knowledge graphs and memory-enhanced recurrent reasoning network.

Table 2

State-of-the-art comparison on Visual7W + KB dataset.

Method	Overall accuracy	
	top-1	top-3
KDMN-NoKnowledge [20]	45.1	-
KDMN-NoMemory [20]	51.9	-
KDMN [20]	57.9	-
KDMN-Ensemble [20]	60.9	-
Out of the Box (OB) [6]	57.32	71.61
GRUC (ours)	69.03	88.12

Table 3

State-of-the-art comparison on OK-VQA dataset. We show the results for the full OK-VQA dataset and for each knowledge category (top-1 accuracy): Vehicles and Transportation (VT); Brands, Companies and Products (BCP); Objects, Material and Clothing (OMC); Sports and Recreation (SR); Cooking and Food (CF); Geography, History, Language and Culture (GHLC); People and Everyday Life (PEL); Plants and Animals (PA); Science and Technology (ST); Weather and Climate (WC); and Other.

Method	Overall accuracy		VT	BCP	OMC	SR	CF	GHLC	PEL	PA	ST	WC	Other
	top-1	top-3											
Q-Only [21]	14.93	-	14.64	14.19	11.78	15.94	16.92	11.91	14.02	14.28	19.76	25.74	13.51
MLP [21]	20.67	-	21.33	15.81	17.76	24.69	21.81	11.91	17.15	21.33	19.29	29.92	19.81
BAN [44]	25.17	-	23.79	17.67	22.43	30.58	27.90	25.96	20.33	25.60	20.95	40.16	22.46
MUTAN [45]	26.41	-	25.36	18.95	24.02	33.23	27.73	17.59	20.09	30.44	20.48	39.38	22.46
ArticleNet (AN) [21]	5.28	-	4.48	0.93	5.09	5.11	5.69	6.24	3.13	6.95	5.00	9.92	5.33
BAN + AN [21]	25.61	-	24.45	19.88	21.59	30.79	29.12	20.57	21.54	26.42	27.14	38.29	22.16
MUTAN + AN [21]	27.84	-	25.56	23.95	26.87	33.44	29.94	20.71	25.05	29.70	24.76	39.84	23.62
BAN/AN oracle [21]	27.59	-	26.35	18.26	24.35	33.12	30.46	28.51	21.54	28.79	24.52	41.4	25.07
MUTAN/AN oracle [21]	28.47	-	27.28	19.53	25.28	35.13	30.53	21.56	21.68	32.16	24.76	41.4	24.85
GRUC (ours)	29.87	32.65	29.84	25.23	30.61	30.92	28.01	26.24	29.21	31.27	27.85	38.01	26.21

异常结果分析

One of our model on OK-VQA is not that the performance on FVQA and Visual7W-KB, we believe that this phenomenon is mostly due to the following two reasons: (1) Questions in the OK-VQA dataset are more diverse and complex, which is more challenging for machines to understand accurately. The questions in FVQA and Visual7W-KB are generated when given the images and supporting facts upon the pre-defined templates or relations. This mechanism constrains the answers in a specific knowledge base and guides the model to operate in a reverse way of the question generation process to predict the correct answers with high probability. On the contrary, questions in OK-VQA are totally free-form ones that generated by MTurk workers and thus containing more unique questions and words with less bias compared with other datasets. This increases the difficulty to understand the questions accurately. (2) OK-VQA requires a wide range of knowledge beyond a specific knowledge base. Looking at the category breakdowns in Table 3, baseline models achieve relatively high performance for SR, CF, GHLC, PA and WC categories while our model performs better for the remaining categories. Since the baseline models refer to the Wikipedia while our model refers to ConceptNet, the performance in the category breakdowns perhaps suggests that each knowledge



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不同维度对比分析

Ablation

TABLE IV

ABLATION STUDY ON VALIDATION SET OF VISDIAL v1.0.

Model	MRR	R@1	R@5	R@10	Mean	NDCG
ObjRep	63.84	49.83	81.27	90.29	4.07	55.48
RelRep	63.63	49.25	81.01	90.34	4.07	55.12
VisNoRel	63.97	49.87	81.74	90.60	4.00	56.73
VisMod	64.11	50.04	81.78	90.52	3.99	56.67
GlCap	60.02	45.34	77.66	87.27	4.78	50.04
LoCap	60.95	46.43	78.45	88.17	4.62	51.72
SemMod	61.07	46.69	78.56	88.09	4.59	51.10
w/o ELMo	63.67	49.89	80.44	89.84	4.14	56.41
		50.74	82.10	91.00	3.91	57.30
		50.79	82.41	91.10	3.90	58.24

Variants介绍

B. Ablation Study

We also conduct an ablation study to further exploit the influence of the essential components of DualVD. To be mentioned, we use DualVD-LF as the full model and apply the same discriminative decoder for all the following variations:

Object Representation (ObjRep): this model uses the averaged object features to represent the image. Question-driven attention is applied to enhance the object representations.

Relation Representation (RelRep): this model applies averaged relation-aware object representations as the image representation without fusing with original object features.

Vision Module without Relationships (VisNoRel): this model contains the full Vision Module, differing in that the relation embeddings are replaced by unlabeled edges.

Visual Module (VisMod): this is our full visual module, which fuses objects and relation features.

Global Caption (GlCap): this model uses LSTM to encode the global caption to represent the image.

Local Caption (LoCap): this model uses LSTM to encode the local captions to represent the image.

Semantic Module (SemMod): this is our full semantic module, which fuses global and local features.

w/o ELMo: this is our full model based on late fusion encoder, differing in that the word embedding GloVe+ELMo is replaced by GloVe.

DualVD-LF (full model): this is our full model, which incorporates both the visual module and semantic module.

Table IV shows the ablation results on VisDial v1.0 validation set. Models in the first block are designed to evaluate the influence of key components in the visual module. The limitation for **ObjRep** is that it only mines the pivotal features from isolated objects and ignores the relational information, which achieves worse performance at all metrics compared to **VisMod**. **RelRep** considers the relationships by introducing relation embedding for aggregating the object features. However, empirical study indicates that enhancing object relationships while weakening object appearance is still not sufficient to represent the visual semantics for better performance. **VisNoRel** takes a step further by adaptively fusing the information from both object appearance and full-connected neighbors, aggregating all the neighborhood features directly without relation semantics. This strategy achieves slight improvement compared with **ObjRep**. **VisMod** moves a step further by adaptively fusing the information from both object appearance and full-connected neighbors, aggregating all the neighborhood features with relational information, which achieves the best performance compared to above two models.

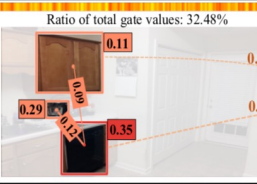
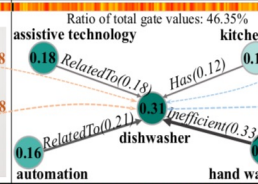
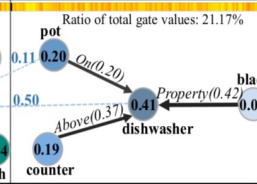
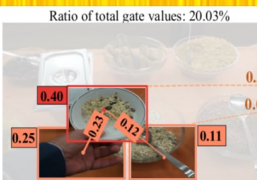
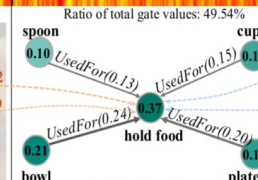
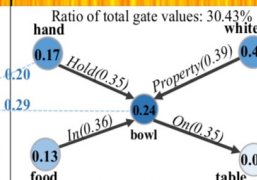
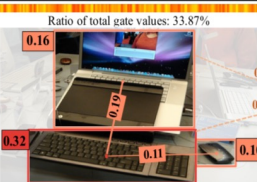
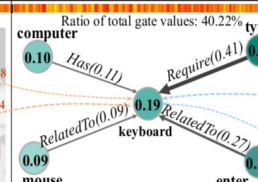
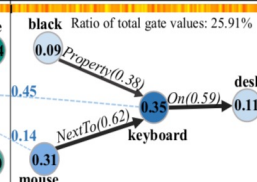
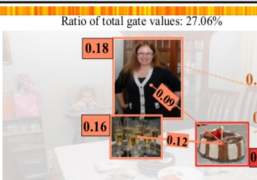
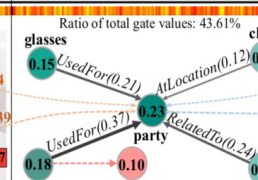
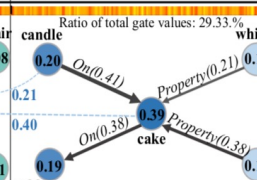
Orthogonal to visual part, models in the second block are conducted to test the influence of key components in the semantic part. The overall performance of either **GlCap** or **LoCap** decreases by 1% and 0.15% respectively, compared to their integrated version **SemMod**, which adaptively selects and fuses the task-specific descriptive clues from both global-level and local-level captions.

We compare the performance of **VisMod** and **SemMod** with **DualVD-LF**. By adaptively select information from the visual and the semantic module, **DualVD-LF** results in a great boost compared to **SemMod** and a relatively slight boost compared to **VisMod**. This unbalanced boost indicates that visual module provides comparatively richer clues than semantic module. Combining the two modules together gains an extra boost, because of the complementary information derived from different modalities. By paying more attention on



一篇论文的组成——实验

Interpretation

Case	Visual Graph	Fact Graph	Semantic Graph
 Question: Which device in the image can free peoples hand? Pred./ Gt Answer: dishwasher (✓)	 Ratio of total gate values: 32.48%	 Ratio of total gate values: 46.35%	 Ratio of total gate values: 21.17%
 Question: What is the white round thing held by hand in the image used for? Pred./Gt Answer: hold food (✓)	 Ratio of total gate values: 20.03%	 Ratio of total gate values: 49.54%	 Ratio of total gate values: 30.43%
 Question: which part of the machine in the image can be used for typing? Pred./GT Answer: keyboard (✓) OB Answer: laptop (X)	 Ratio of total gate values: 33.87%	 Ratio of total gate values: 40.22%	 Ratio of total gate values: 25.91%
 Question: Where can you find the right object on the table shown in the image? GT Answer: wedding Pred. Answer: party (X)	 Ratio of total gate values: 27.06%	 Ratio of total gate values: 43.61%	 Ratio of total gate values: 29.33%

Our model is interpretable by visualizing the attention weights and gate values in the reasoning process. From case study in Figure 3, we conclude with the following three insights: (1) **Mucko is capable to reveal the knowledge selection mode.** The first two examples indicate that Mucko captures the most relevant visual, semantic and factual evidence as well as complementary information across three modalities. In most cases, factual knowledge provides predominant clues compared with other modalities according to gate values because FVQA relies on external knowledge to a great extent. Furthermore, more evidence comes from the semantic modality when the question involves complex relationships. For instance, the second question involving the relationship between ‘hand’ and ‘while round thing’ needs more semantic clues. (2) **Mucko has advantages over the state-of-the-art model.** The third example compares the predicted answer of OB with Mucko. Mucko collects relevant visual and semantic evidence to make each entity discriminative enough for predicting the correct answer while OB failing to distinguish representations of ‘laptop’ and ‘keyboard’ without feature selection. (3) **Mucko fails when multiple answers are reasonable for the same question.** Since both ‘wedding’ and



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Parameter Analysis

#Retrieved facts	@50	@100	@150	@200
Rel@1 (top-1 accuracy)	55.56	70.62	65.94	59.77
Rel@1 (top-3 accuracy)	64.09	81.95	73.41	66.32
Rel@3 (top-1 accuracy)	58.93	73.06	70.12	65.93
Rel@3 (top-3 accuracy)	68.50	85.94	81.43	74.87

Table 3: Overall accuracy with different number of retrieved candidate facts and different number of relation types.

#Steps	1	2	3
top-1 accuracy	62.05	73.06	70.43
top-3 accuracy	71.87	85.94	81.32

Table 4: Overall accuracy with different number of reasoning steps.

其他需要验证的实验

- 泛化能力（更多不同任务）
- 局部模块效果验证
- 存在的局限性
- ...



一篇论文的组成——结论

基本要求

- ☀ 和摘要的区别：经过验证后的结论
- ☀ 论文主要发现总结
- ☀ 介绍小问题的结论
- ☀ 避免过度夸大
- ☀ 内容尽量简洁
- ☀ 介绍未来工作



一篇论文的组成——结论

Abstract

Fact-based Visual Question Answering (FVQA) requires external knowledge beyond visible content to answer questions about an image, which is challenging but indispensable to achieve general VQA. One limitation of existing FVQA solutions is that they jointly embed all kinds of information without fine-grained selection, which introduces unexpected noises for reasoning the final answer. How to capture the question-oriented and information-complementary evidence remains a key challenge to solve the problem. In this paper, we depict an image by a multi-modal heterogeneous graph, which contains multiple layers of information corresponding to the visual, semantic and factual features. On top of the multi-layer graph representations, we propose a modality-aware heterogeneous graph convolutional network to capture evidence from different layers that is most relevant to the given question. Specifically, the intra-modal graph convolution selects evidence from each modality and cross-modal graph convolution aggregates relevant information across different modalities. By stacking this process multiple times, our model performs iterative reasoning and predicts the optimal answer by analyzing all question-oriented evidence. We achieve a new state-of-the-art performance on the FVQA task and demonstrate the effectiveness and interpretability of our model with extensive experiments. The code is available at <https://github.com/lestra-zibao/mucko>.

说明动机
陈述方法

Conclusion

In this paper, we propose Mucko for visual question answering requiring external knowledge, which focuses on multi-layer cross-modal knowledge reasoning. We novelly depict an image by a heterogeneous graph with multiple layers of information corresponding to visual, semantic and factual modalities. We propose a modality-aware heterogeneous graph convolutional network to select and gather intra-modal and cross-modal evidence iteratively. Our model outperforms the state-of-the-art approaches remarkably and obtains interpretable results on the benchmark dataset.

总结工作
体现效果

ation. However, our model has inferior performance when open-domain knowledge is required. How to adaptively incorporate diverse knowledge bases that covering commonsense, Wikipedia knowledge and even professional knowledge for KVQA tasks will be our future work.

说明局限
指明方向



一篇论文的组成——致谢

基本要求

☀ 帮助这篇论文的人员、机构、项目资助

☀ 审稿人

☀ 提供建议的其他科研人员

☀ 非*co-author*



一篇论文的组成——参考文献

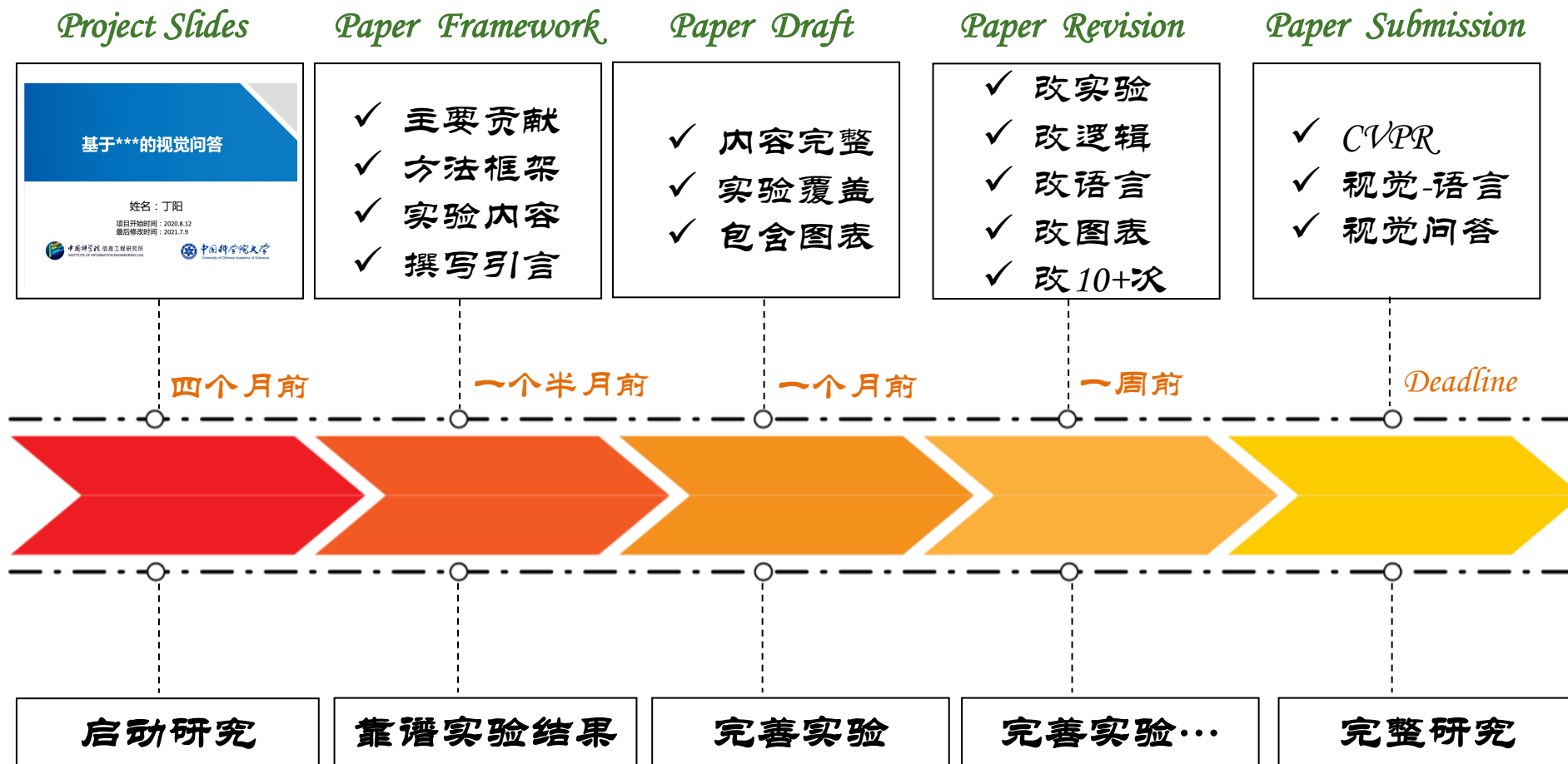
基本要求

☀ 不遗漏，查全

☀ 按照会议/期刊既定格式

☀ 常见错误：大小写、全称/缩写、漏写、名字错拼

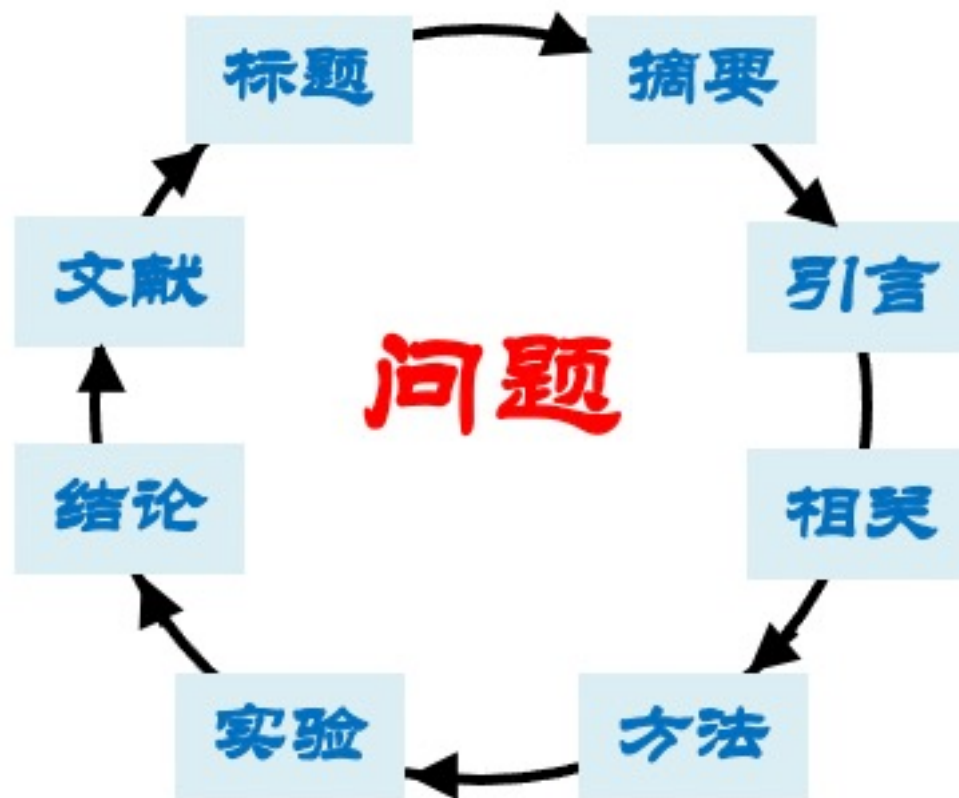
论文写作及修改过程



Take Home Message

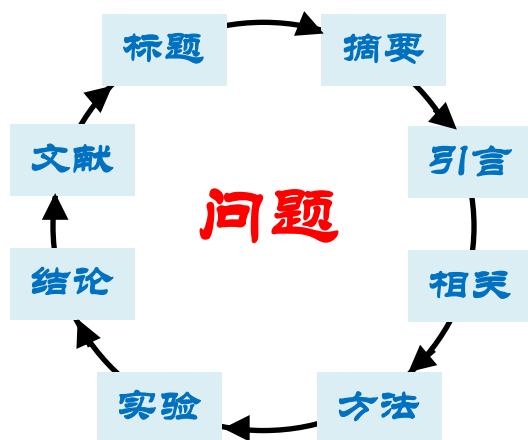


写作思路

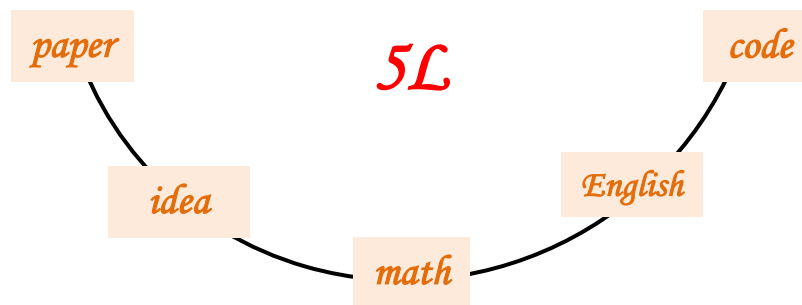
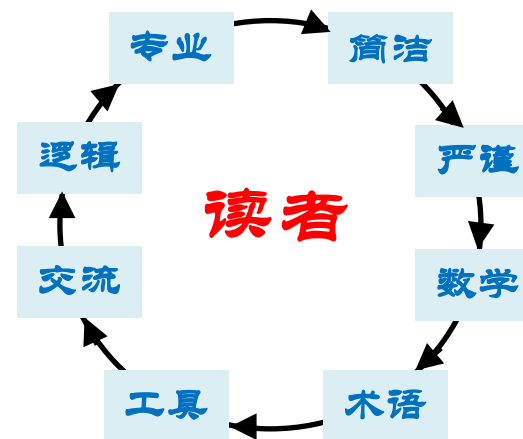


Take Home Message

☀️ 写作思路



☀️ 英文规范



☀️ 日常积累



参考书籍

- ☀ Booth, Wayne C., et al. *The craft of research*. University of Chicago press, 2003.
- ☀ Petre, Marian, and Gordon Rugg. *The Unwritten Rules Of Phd Research*. McGraw-Hill Education (UK), 2010.
- ☀ Macrina, Francis L. *Scientific Integrity*. ASM Press, 2005.

参考课程

- ☀ [董彬，北京大学，北京国际数学研究中心，《研究型学习》](#)

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对本报告内容的指导和启发！**

*As long as I am dreaming, believing and doing,
I can go anywhere and achieve anything!*

于静

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研究组主页



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